

**GRAPH THEORY:
HOMEWORK 6
BOOK SECTION 3.1 PART II.**

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1. Prove or disprove: Every tree has at most one perfect matching. (Hint: If a tree had two different perfect matchings, what could you say about the edges that belong to exactly one of them?)

Proof. □

2. Prove that every maximal matching in a graph G has at least $\alpha'(G)/2$ edges. (Hint: Compare a maximal matching M with a maximum matching M^* . What does maximality of M tell you about each edge of the maximum matching?)

Proof. □

3. Prove that if G is an X, Y -bigraph such that

$$|N(S)| > |S| \quad \text{whenever } \emptyset \neq S \subsetneq X,$$

then every edge of G belongs to some matching that saturates X . (Hint: Given an edge xy , consider the graph $G' = G - x - y$. What would you like to show about G' , and how does deleting y affect neighborhoods?)

Proof. □

4. Use the König–Egerváry Theorem to prove that every bipartite graph G has a matching of size at least $e(G)/\Delta(G)$. Use this to conclude that every subgraph of $K_{n,n}$ with more than $(k-1)n$ edges has a matching of size at least k . (Hint: König–Egerváry lets you bound a different parameter instead of $\alpha'(G)$. How many edges can a single vertex take care of?)

Proof. □

5. Let G be an X, Y -bigraph with vertex cover Q , and let H be the induced $(X-Q), (Q \cap Y)$ -sub-bigraph of G . Prove that if no matching in H saturates $Q \cap Y = Q - X$, then G has a vertex cover that is smaller than Q . (Hint: What does Hall's Theorem tell you when no matching in H saturates $Q - X$? Use that to decide which vertices of Q you might remove and replace to obtain a smaller set, and explain why the new set is a vertex cover.)

Proof. □